Carnatic Classical Raga Identification Using Stacked Ensemble Learning Model

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***Abstract*— This paper presents an innovative approach for Raga Identification in Carnatic Classical Music, leveraging a Stacked Ensemble Learning Model alongside individual classifiers, namely Multi-Layer Perceptron (MLP), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest (RF). The dataset encompasses diverse ragas, both Melakartha and Janya, with features like Mel-Frequency Cepstral Coefficient(MFCC), Spectral Centroid, and Chroma Short Time Fourier Transform. The individual classifiers, providing accuracies 96.718% for KNN, 96.878% for MLP, 96.875% for RF, and 97.031% for SVM, demonstrate commendable performance. The Stacked ensemble mode however outperforms individual classifiers, achieving an accuracy of 97.812%. Result analysis incorporates precision, recall, F1 measure, accuracy, and support metrics, illustrating the ensemble's superior performance in capturing the intricate nuances of ragas.**

This paper not only contributes to the burgeoning field of computational music analysis but also sheds light on the efficacy of ensemble learning for nuanced Raga Identification. The findings underscore the potential of the proposed model to accurately classify diverse ragas, paving the way for advancements in automated recognition systems for Indian Classical Music. Future work may explore hyperparameter tuning and the generalization of the proposed model across varied datasets and musical renditions.

Keywords—Raga Identification, Carnatic Classical Music, Computational Music Analysis, Ensemble Learning, Stacked Ensemble Model, Multi-Layer Perceptron, Support Vector Machine, K-Nearest Neighbors, Random Forest, Melakartha Ragas, Janya Ragas, Chroma Short-Time Fourier Transform, Spectral Centroid, Mel-Frequency Cepstral Coefficients.

I. INTRODUCTION

Indian Classical Music, with its rich heritage and diverse melodic structures, has been a subject of interest for researchers exploring computational techniques for music analysis and classification. Raga identification, a crucial aspect of Indian Classical Music, involves recognizing and categorizing the intricate melodic patterns that define specific ragas[1]. Over the years, various methodologies have been employed to automate the process of raga identification, ranging from traditional clustering techniques to cutting-edge deep learning models.

The exploration of machine learning techniques for raga classification has seen significant progress, with several studies delving into innovative approaches. In the work by Vasudevan et al. [2], a "Hybrid Cluster-Classifier model" was proposed, combining clustering techniques with classification models to enhance Carnatic raga classification. This approach demonstrated promising results, laying the foundation for further advancements in computational music analysis.

Deep learning has also emerged as a powerful tool for pattern recognition tasks, and its application to Indian Classical Music has not gone unnoticed. The paper by Shah et al. [3] leveraged deep learning techniques for Raga Recognition, showcasing the ability of neural networks to discern intricate melodic nuances.

A comprehensive survey conducted by Waghmare et al. [4] explored various Raga Identification Techniques, providing an overview of the existing landscape of methodologies and their comparative merits. This survey serves as a guide, highlighting the evolution of techniques employed for classifying Indian Classical Music based on ragas.

Furthermore, the integration of deep learning in Singha et al.'s work [5] also demonstrated the potential for automated classification, opening new avenues for leveraging neural networks in the discipline of Indian Classical Music. This underscores the importance of latest advancements in machine learning for enhanced accuracy and efficiency.

In this context, our paper addresses the task of Raga Identification through the novel application of a Stacked Ensemble Learning Model. Our ensemble model leverages the strengths of base classifiers, including Multi-Layer Perceptron (MLP), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest (RF), due to their popularity as classifiers and available literature of music classifications. The utilization of a Random Forest meta-learning model enhances the ensemble's overall performance, providing a robust and accurate solution for raga identification.

This paper hence adds to the ongoing discourse on computational music analysis by presenting a unique approach to Raga Identification. The utilization of a stacked ensemble model, informed by insights from previous works, represents a significant stride toward achieving more

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accurate and nuanced results in the sphere of Indian Classical Music classification.

II. Description of Dataset

The dataset exercised in this study for Raga Classification was acquired from "Kaggle" and sourced from "Ragas with features in Indian classical music" represented in a CSV file [6]. The dataset was meticulously curated by using popular ragas, with each 30-second audio file containing recordings of distinct ragas. The CSV file, encompasses a total of 1112 instances, each characterized by 26 features. There are total of 32 Carnatic raga classes covering some of the popular Melakartha and Janya ragas in Carnatic Classical Music. The dataset serves as a foundation for evaluating the performance of various classifiers, including Multi-Layer Perceptron, Support Vector Machine, K-Nearest Neighbors, and Random Forest and a Stacked Ensemble Learning Model.

Melakartha and Janya Ragas:

Melakartha ragas, also known as Janaka ragas, form the fundamental parent ragas in the Carnatic Classical Music system. Melakartha in Carnatic music is equivalent of Thaat in Hindustani Classical Music[7]. There are precisely 72 Melakartha ragas, each encompassing seven swaras or musical notes. Among the ragas used in this dataset, seven are Melakartha ragas, as in Kalyani, Harikambhoji, Karaharapriya, Mayamalavagaula, Charukesi, Shankarabharam and Todi.

Janya ragas, also known as derivative ragas, are created by deriving musical scales from one of the Melakartha ragas. There can be numerous Janya ragas associated with each Melakartha raga, resulting in a vast and diverse repertoire of ragas in Carnatic Classical Music. In the dataset, the majority of the ragas belong to the category of Janya ragas.

Data Preprocessing:

The CSV file captures various audio features, including Mel-Frequency Cepstral Coefficient(MFCC), Root Mean Square Error (RMSE), Spectral Centroid, and Chroma Short Time Fourier Transform, and Spectral Bandwidth in addition to the corresponding raga label[8]. Many of the data features proved to be very essential for classification purpose while some features were eliminated due to redundancy and minimal feature importance.

1. Data Cleaning: Firstly the unnecessary columns such as filename and file number are removed for each audio file. RMSE (Root Mean Square Error) is next eliminated due to its potential unsuitability for capturing the distinctive features relevant to this task. While RMSE is mostly relevant to regression tasks, our focus in this paper remains completely upon multiple raga classification.
The column of Spectral bandwidth is ignored as well due to its redundancy with other features for this particular case and to streamline the dataset. MFCC5 (Mel-Frequency Cepstral Coefficient-5) was also discarded as it had the least feature importance compared to other coefficients. This proved beneficial for overall dimensionality reduction and prevented unnecessary overfitting.
This finally leaves us with 20 raga features and a raga label.

2. Class Balancing: Given the potential class imbalance in the dataset, Synthetic Minority Over-sampling Technique (SMOTE) is employed to fabricate synthetic instances of the minority classes. This aids in creating a more balanced representation of the ragas, preventing bias towards the majority classes during model training.

The various ragas used in this dataset along with their swara sequences and types are depicted in Table. 1.

Features:

1. Chroma Short-Time Fourier Transform (chroma_stft): Chroma Short-Time Fourier Transform is a representation of the audio signal in the frequency domain. It calculates the energy distribution of pitches within the 12 different pitch classes, effectively summarizing the presence of musical notes in the audio.
Chroma features are widely used in music information retrieval tasks, including genre classification and raga identification. The chroma representation is particularly valuable for capturing tonal characteristics and harmonic content, making it applicable for discerning distinctive melodic patterns in Indian Classical Music[9].

2. Spectral Centroid (spec_cent):
Spectral Centroid is a measure of the "center of mass" of the spectrum, indicating where the "center" of the audio signal's frequency distribution lies. It provides insights into the overall tonal characteristics of the sound, identifying whether it is more biased towards higher or lower frequencies.
In the context of raga classification, it aids in capturing the dominant frequency components, offering valuable information about the tonal center of the audio signal and contributing to the characterization of specific ragas.

3. Mel-Frequency Cepstral Coefficients(mfcc0 - mfcc18): Mel-Frequency Cepstral Coefficients capture the short- term power spectrum of an audio signal, mimicking the human auditory system's sensitivity to different frequencies. The coefficients are derived from the spectrum after it undergoes a series of processing steps, including mel- frequency scaling and logarithmic compression.
MFCCs are fundamental features in audio signal processing and are extensively used for various tasks, including speech and music analysis. In the context of raga classification, MFCCs capture essential timbral and spectral characteristics, enabling the model to discern subtle differences in tonal quality that are distinctive to different ragas[10].

These features collectively contribute to the task of raga classification by capturing critical aspects of the audio signal. Chroma features excel at representing melodic content, while Spectral Centroid provides insights into the tonal center. MFCCs, being comprehensive in nature, encapsulate both tonal and timbral aspects, making them

particularly valuable for distinguishing the unique characteristics of different ragas[11][12].
The combination of these features in the dataset facilitates the training of machine learning models, enabling them to learn and recognize patterns inherent to specific ragas.

Table.1 Various ragas used in the dataset

No.	RAGA	RAGA TYPE	Swara Sequences
1	Kapi	Janya	S R2 M1 P N3 S
2	Hamsanadam	Janya	S R2 M2 P N3 S
3	Bhairavi	Janya	S G2 R2 G2 M1 P D2 N2 S
4	Jaganmohini	Janya	S G3 M1 P N3 S
5	Hamsadhwani	Janya	S R2 G3 P N3 S
6	Kalyani	Melakartha	S R2 G3 M2 P D2 N3 S
7	Harikambhoji	Melakartha	S R2 G3 M1 P D2 N2 S
8	Kamas	Janya	S M1 G3 M1 P D2 N2 S
9	Darbari kannada	Janya	S R2 G2 M1 P D1 N2 S
10	Bilahari	Janya	S R2 G3 P D2 S
11	Kanada	Janya	S R2 P G2 M1 D2 N2 S
12	Gaula	Janya	S R1 M1 P N3 S
13	Sindhubhairavi	Janya	S R1 G2 M1 P D1 N2 S
14	Ranjani	Janya	S R2 G2 M2 D2 S
15	Kedaram	Janya	S M1 G3 M1 P N3 S
16	Sahana	Janya	S R2 G3 M1 P M1 D2 N2 S
17	Mohanam	Janya	S R2 G3 P D2 S
18	Karaharapriya	Melakartha	S R2 G2 M1 P D2 N2 S
19	Mayamalavagaula	Janya	S R1 G3 M1 P D1 N3 S
20	Varali	Janya	S G1 R1 G1 M2 P D1 N3 S
21	Amritavarshini	Janya	S G3 M2 P N3 S
22	Anandabhairavi	Janya	S G2 R2 G2 M1 P D2 P S
23	Nata	Janya	S R3 G3 M1 P D3 N3 S
24	Charukesi	Melakartha	S R2 G3 M1 P D1 N2 S
25	Shankarabharam	Melakartha	S R2 G3 M1 P D2 N3 S
26	Saveri	Janya	S R1 M1 P D1 S
27	Todi	Melakartha	S R1 G2 M1 P D1 N2 S
28	Kamboji	Janya	S R2 G3 M1 P D2 S
29	Madhyamavati	Janya	S R2 M1 P N2 S
30	Sriranjini	Janya	S R2 G2 M1 D2 N2 S
31	Hindolam	Janya	S G2 M1 D1 N2 S
32	Janaranjani	Janya	S R2 G3 M1 P D2 P N3 S

III. METHODOLOGY

After proper preprocessing, the data is fed into various machine learning models for the task of classification. The overall data-flow is illustrated in a flowchart represented in Fig. 1.

The methodology employed in this study leverages a blend of foundational machine learning models and a Stacked Ensemble Learning Model to achieve accurate and nuanced Raga Identification in Indian Classical Music.

The dataset includes 32 classes, representing different ragas, and incorporates a diverse range of Melakartha and Janya ragas, adding depth and complexity to the classification task. The dataset is allocated into training and testing partitions, adhering to an 80:20 distribution, ensuring a robust evaluation of model performance on unseen data[13][14].

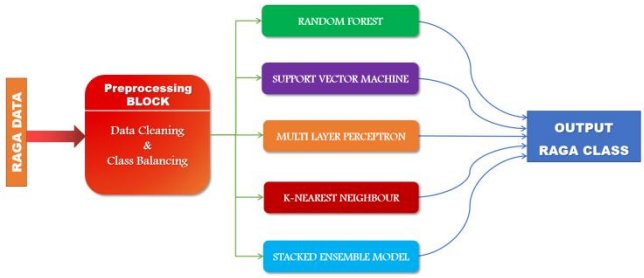

Fig.1 Raga Identification Pipeline

Base Models:

1. Random Forest (RF): A decision tree-based ensemble model known for its robustness and ability to handle diverse datasets. RF is utilized to capture complex relationships within the features, contributing to the overall predictive power. Here the model utilizes 100 estimators.

2. Support Vector Machine (SVM): Employing the principles of hyper-plane separation, SVM is effective in distinguishing between different classes in high-dimensional spaces. It is used to capture intricate boundaries between ragas.

3. Multi-Layer Perceptron (MLP): A neural network model competent of learning intricate patterns and relationships. MLP is applied to capture non-linear relationships within the dataset, enhancing the model's ability to discern subtle melodic nuances. The model used here consists of three hidden layers with sizes 256, 128, and 64.

4. K-Nearest Neighbor (KNN): A proximity-based model, KNN classifies instances assessing the majority class among their nearest neighbors. It is utilized to capture localized relationships within the dataset. Here the designed model associates the each data point with three neighbors.

Stacked Ensemble Model

The Stacked Ensemble Model makes a potent ensemble learning technique that amalgamates predictions from multiple base models, elevating the overall predictive performance[15]. The ensemble model here, utilizes a Random Forest meta-model to make final predictions.

The crucial stages in the workflow are-
Step 1: Base models (RF, SVM, MLP, KNN) are trained individually on the dataset, each capturing unique aspects of the data.
Step 2: The forecasts generated by the base models are subsequently employed as inputs for the Random Forest meta-model.
Step 3: The meta-model learns to combine these predictions, effectively leveraging the diversity of the base models to enhance overall accuracy and generalization.

The Stacked Ensemble learning model is depicted in Fig. 2.

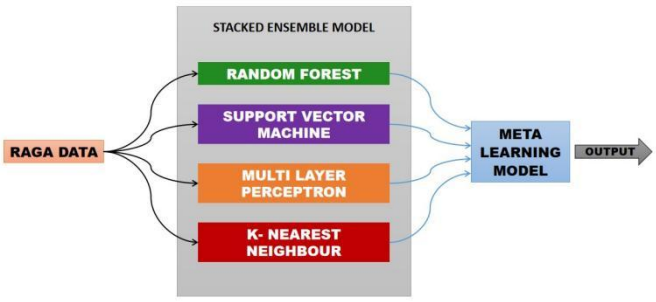

Fig.2 Stacked Ensemble learning model

IV. Result Analysis And Discussion

In the pursuit of evaluating the efficacy of the suggested base models and the Raga Identification system utilizing Stacked Ensemble Learning Model, a comprehensive result analysis has been conducted. The performance of all individual classifiers, namely KNN, MLP, RF, and SVM, has been compared with the Stacked ensemble model.

Confusion matrices for the same were generated for a detailed class-wise analysis of each of the 32 raga classes. This provided a comprehensive insight into the strengths and weaknesses of each classifier, illuminating their proficiency in distinguishing among the intricate nuances of the 32 Carnatic classical ragas. The confusion matrices also facilitated a granular examination of classification outcomes, offering an exhaustive analysis of true positives, true negatives, false positives, and false negatives for each raga. The confusion matrices for RF, SVM, MLP, KNN and Stacked Ensemble models have been depicted in Fig. 3, Fig. 4, Fig. 5, Fig. 6 and Fig. 7 respectively.

Moreover precision, recall, F1 measure, accuracy, and support metrics were also computed to evaluate the comprehensive performance of the classification model.

Precision: Precision measurement of the accuracy of the positive predictions. It is estimated as the ratio of true positive predictions to the total of true positives and false positives. Higher precision indicates fewer false positives.
Recall: Recall, also known as sensitivity or true positive rate, gauges the ability of the model to capture all relevant instances. It is computed as the ratio of true positive predictions to the total of true positives and false negatives.
F1 Measure: The F1 measure is the harmonic mean of precision and recall. It provides a balanced assessment of a classifier's performance, especially in scenarios where imbalances exist between positive and negative instances.
Accuracy: Accuracy measures the overall veracity of predictions and is calculated as the ratio of correct predictions to the total number of predictions.
Support: Support refers to the number of actual occurrences of each class in the dataset.

The various performance metrics for each model have been illustrated in Table 2.

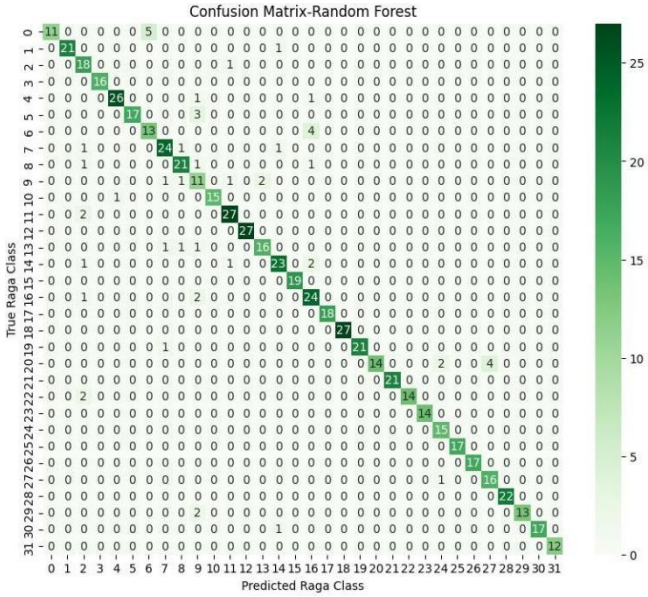

Fig. 3 Confusion matrix for RF

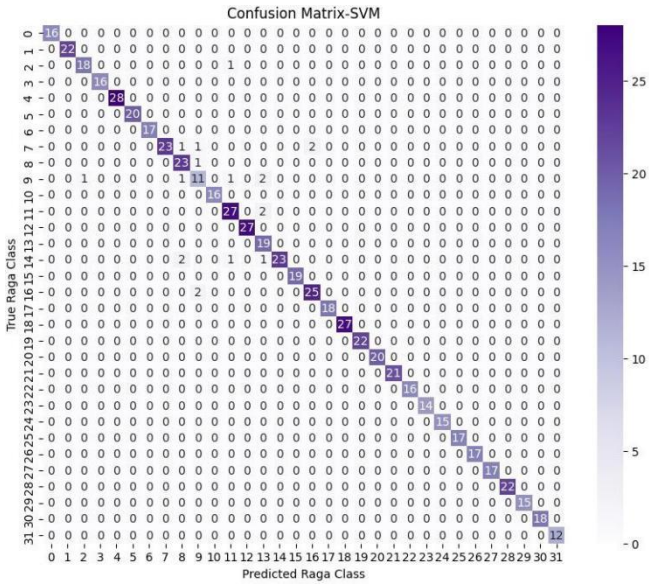

Fig. 4 Confusion matrix for SVM

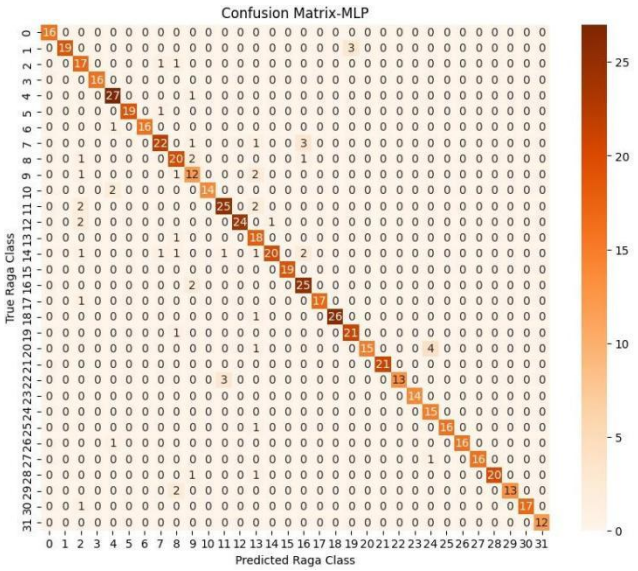

Fig. 5 Confusion matrix for MLP

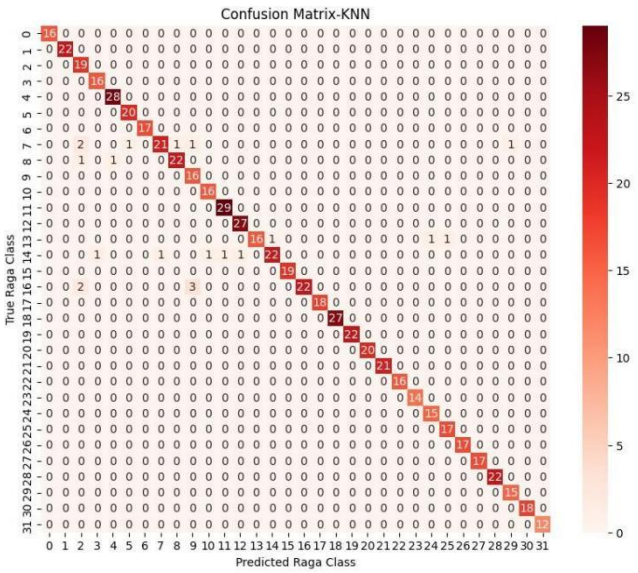

Fig. 6 Confusion matrix for KNN

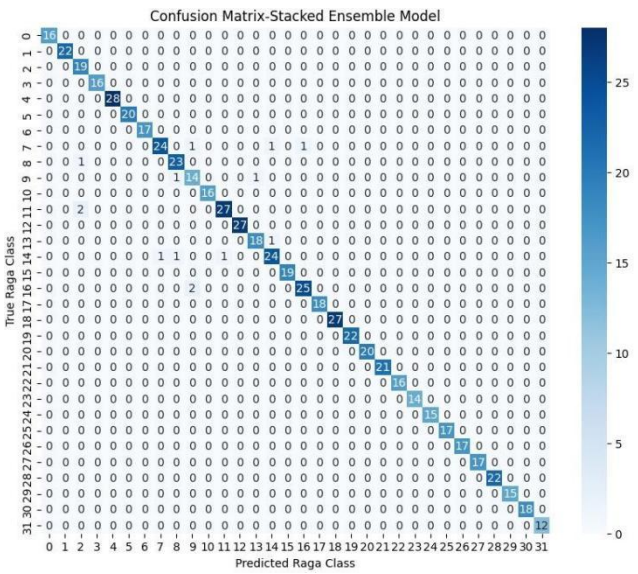

Fig. 7 Confusion matrix for Stacked Ensemble Model

Table. 2 Performance metrics for various classifiers

ML Model	Precision	Recall	f1-score	Support	Accuracy
RF	0.97	0.97	0.97	640	96.875
SVM	0.97	0.97	0.97	1456	97.031
MLP	0.97	0.97	0.97	640	96.878
KNN	0.97	0.97	0.97	640	96.718
Stacked Model	0.98	0.98	0.98	640	97.812

The obtained accuracies showcase the competence of each individual classifier in capturing the intricacies of Raga Identification. Notably, the Raga-Stacked model outperforms its individual counterparts, indicating the efficacy of the ensemble learning approach in enhancing classification accuracy. The Raga-Stacked model exhibits higher accuracy (97.812%) compared to individual classifiers, indicating a collective learning approach is beneficial in the context of Raga Identification.

The notable accuracy improvement with the Raga-Stacked model underscores the potential of ensemble learning in enhancing the classification of Indian Classical ragas. The robustness of the proposed system can be assessed by introducing diverse datasets, representing variations in recording conditions and musical renditions.

A bar graph plot for accuracy comparison is provided in Fig. 8.

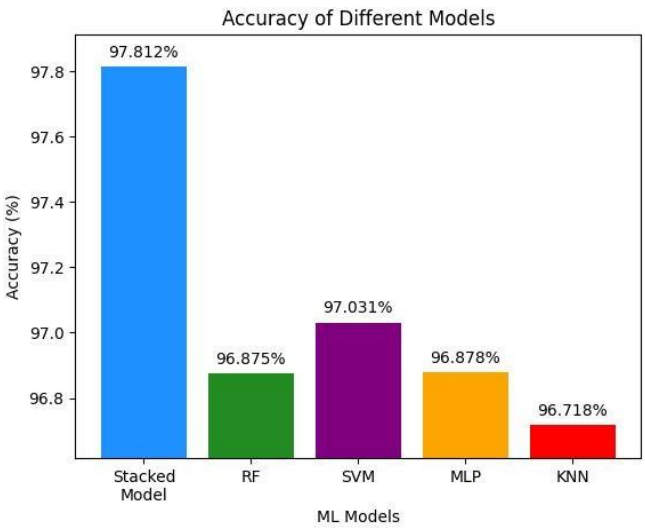

Fig.8 Bar graph for accuracy comparison

V. CONCLUSION

In conclusion, this paper introduces an advanced framework for Raga Identification in Carnatic Classical Music, combining the strengths of individual classifiers and a Stacked Ensemble Learning Model. The Stacked ensemble model, emerges as a robust solution, outperforming individual classifiers, including RF, SVM, MLP, and KNN. The ensemble model achieves an overall impressive accuracy of 97.812%, showcasing its efficacy in capturing the diverse melodic patterns inherent to ragas.

The comprehensive result analysis, incorporating precision, recall, F1 measure, accuracy, and support metrics, highlights the ensemble's ability to provide a nuanced and balanced classification. The individual classifiers also demonstrate commendable accuracy, emphasizing the significance of a diverse set of models in the context of Raga Identification.

The success of the proposed framework for Raga Identification using ensemble learning opens avenues for future research and possible upgrades in the arena of computational music analysis and development of automated systems capable of recognizing and classifying diverse ragas in Indian Classical Music. Several other promising directions and potential areas for improvement and exploration may include-- Transfer Learning and Pre-trained Models, Real-time Raga Recognition, Cultural and Regional Raga Variations and many other Raga based applications in the Classical Music Industry.

REFERENCES

[1] Padmasundari, "Music Information Retrieval: Raga Identification using Machine Learning", MA Thesis, Industrial Math. and Sci. Computing, IITM, Chennai, India, 2016.

[2] A. Vasudevan and H. Ramasangu, "A Hybrid Cluster-Classifier model for Carnatic Raga Classification", IEEE Int. Conf. on Electro., Computing and Comm. Tech. (CONECCT), Bangalore, India, 2021, pp. 1-6, doi: 10.1109/CONECCT52877.2021.9622669.

[3] D. Shah, N. M. Jagtap, P. T. Talekar, and K. Gawande, "Raga Recognition in Indian Classical Music Using Deep Learning", AI in Music, Sound, Art and Design, Mumbai, India, 2021, pp. 248-263, doi:10.1007/978-3-030-72914-1_17.

[4] K. C. Waghmare and B. A. Sonkamble, "Raga Identification Techniques for Classifying Indian Classical Music: A Survey", Int. J. of Sig. Process. Sys., Pune, India, 2017, pp. 130-135, doi:10.18178/ijsps.5.4.130-135.

[5] A. Singha, N. R. Rajalakshmi, J. A. Pandian, and S. Saravanan, "Deep Learning-Based Classification of Indian Classical Music Based on Raga", 6th Int. Conf. on Info. Sys. and Comp. Net. (ISCON), Mathura, India, 2023, pp. 1-7, doi: 10.1109/ISCON57294.2023.10111985.

[6] S. Satish, "Ragas with features in Indian classical music", Kaggle, 2023.

[7] A. Mhatre, "Classification of Indian Classical Music Ragas using Graph Theory", Second Int. Conf. on Next Gen. Intelligent Sys. (ICNGIS), Kottayam, India, 2022, pp. 1-5, doi: 10.1109/ICNGIS54955.2022.10079833.

[8] A. R. Chhetri, K. Kumar, M. P. Muthyala, S. M R, and R. A. Bangalore, "Carnatic Music Identification of Melakarta Ragas through Machine and Deep Learning using Audio Signal Processing", 4th Int. Conf. for Emerging Tech. (INCET), Belgaum, India, 2023, pp. 1-5, doi: 10.1109/INCET57972.2023.10170568.

[9] K. C. Waghmare and B. A. Sonkamble, "Analyzing Acoustics of Indian Music Audio Signal Using Timbre and Pitch Features for Raga Identification", 3rd Inter. Conf. on Imaging, Signal Processing and Comm. (ICISPC), Singapore, 2019, pp. 42-46, doi: 10.1109/ICISPC.2019.8935707.

[10] T. Katte and B. S. Tiple, "Techniques for Indian classical raga identification- A survey", Annual IEEE India Conf. (INDICON), Pune, India, 2014, pp. 1-6, doi: 10.1109/INDICON.2014.7030372.

[11] P. Chinthapenta, "Machine Learning for Raga Identification in Indian Classical Music", UG Research Scholars Thesis, Depart. of Elec. and Comp. Engineering, Texas A&M University, Texas, USA, 2019.

[12] H. Sharma, R. S. Bali, "Comparison of ML classifiers for Raga recognition", Int. J. of Sci. and Res. Publications, Chandigarh, India, 2015, Vol. 5, no. 10, ISSN 2250-3153, pp. 1-5, doi: 10.29322.

[13] A. Sridharan, M. Moh, and T. -S. Moh, "Similarity Estimation for Classical Indian Music", 17th IEEE Int. Conf. on ML and Apps. (ICMLA), Orlando, FL, USA, 2018, pp. 814-819, doi: 10.1109/ICMLA.2018.00130.

[14] D. Joshi, J. Pareek, and P. Ambatkar, "Indian Classical Raga Identification using Machine Learning", Int. Symp. on Intelligent Control (ISIC), New Delhi, India, 2021, Vol. 2786, pp. 259-263.

[15] R. Shariat and J. Zhang, "An Empirical Study on the Effectiveness of Feature Selection and Ensemble Learning Techniques for Music Genre Classification", 18th Int. Audio Mostly Conf. (AM '23), New York, USA, 2023, pp. 51–58. doi: 10.1145/3616195.3616217.